National University of Lesotho
Department of Mathematics and Computer Science

M3503 Tutorial 1 August 2025

1. Solve

   (a) $e^x y\dfrac{dy}{dx} = e^{-y} + e^{-2x-y}$

   (b) $x^2\dfrac{dy}{dx} = y - xy, \quad y(-1) = -1$

   (c) $y\ln x\dfrac{dx}{dy} = \left(\dfrac{y+1}{x}\right)^2$

   (d) $\csc(y)dx + \sec^2(x)dy = 0$

   (e) $\sqrt{1-y^2}dx - \sqrt{1-x^2}dy = 0, \quad y(0) = \dfrac{\sqrt{3}}{2}$

   (f) $x^2y' + x(x+2)y = e^x$

   (g) $y' + 3x^2y = x^2$

   (h) $\dfrac{dy}{dx} - y = x^2\sin(x)$

   (i) $(5x+4y)dx + (4x-8y^3)dy = 0$

   (j) $[\sin(y) - y\sin(x)]dx + [\cos(x) + x\cos(y) - y]dy = 0$

   (k) $\left(1 + \ln x + \dfrac{y}{x}\right)dx = (1 - \ln x)dy$

   (l) $(2y^2 + 3x)dx + 2xydy = 0$

   (m) $\cos(x)dx + \left(1 + \dfrac{2}{y}\right)\sin(x)dy = 0$

   (n) $(10 - 6y - e^{-3x})dx - 2dy = 0$

   (o) $(3x^2 - y^2)dx + (xy - x^3y^{-1})dy = 0$

   (p) $\dfrac{dy}{dx} = \dfrac{y^2 + x\sqrt{x^2+y^2}}{xy}$

   (q) $\dfrac{dy}{dx} = (x - y + 5)^2$

   (r) $\dfrac{dy}{dx} = \sin(x - y)$

2. Find a function whose square plus the square of its derivative is 1.

3. According to Newton's law of cooling, if an object at temperature T is immersed in a medium having the constant temperature M, then the rate of change of T is proportional to the difference of temperature $M - T$. This gives the differential equation

$$\frac{dT}{dt} = k(M - T)$$

   A thermometer reading $100\,\text{deg}\,F$ is placed in a medium having a constant temperature of $70\,\text{deg}\,F$. After 6 minutes, the thermometer reads $80\,\text{deg}\,F$. What is the reading after 20 mininutes?

4. Find an integrating factor of the form $x^n y^m$, that will make the given equation exact, and then use it to solve the equation.

   (a) $(2y^2 - 6xy)dx + (3xy - 4x^2)dy = 0$

   (b) $(12 + 5xy)dx + (6xy^{-1} + 3x^2)dy = 0$